fn approximate_to_tradeoff

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This proof resides in **"contrib"** because it has not completed the vetting process.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `epsilon` of type `f64`
- Argument `delta` of type `f64`

Caller-verified

None

Pseudocode

```
def appproximate_to_tradeoff(
    epsilon: f64,
    delta: f64,
) -> Callable[[RBig], RBig]:
    epsilon = FBig.try_from(epsilon, Up)
    delta = RBig.try_from(delta)

    precision = epsilon.precision().max(10)
    epsilon = epsilon.with_precision(precision).value()

    exp_eps = RBig.try_from(epsilon.with_rounding(Up).exp())  #
    exp_neg_eps = RBig.try_from((-epsilon).with_rounding(Down).exp())  #

    def tradeoff(alpha: RBig) -> RBig:  #
        t1 = RBig(1) - delta - exp_eps * alpha
        base = RBig(1) - delta - alpha
        t2 = exp_neg_eps * base
        return max(max(t1, t2), RBig(0))

    return tradeoff
```

Postcondition

Theorem 1.1. The pseudocode returns a function $\tilde{f}$ such that for all $\alpha \in [0, 1]$,

$$\tilde{f}(\alpha) \leq f_{\epsilon,\delta}(\alpha),$$

where $f_{\epsilon,\delta}$ is the corresponding f-DP tradeoff curve.

We start with the following definition of DP from Dong, Roth, and Su 2022.

Definition 1.2 (Dong, Roth, and Su 2022, Proposition 2.5)**.** Let $\epsilon > 0$ and $\delta \geq 0$, and define

$$f_{\epsilon,\delta}(\alpha) = \max(0, 1 - \delta - \exp(\epsilon)\alpha, \exp(-\epsilon)(1 - \delta - \alpha)). \tag{1}$$

Then we say that a mechanism M satisfies (ϵ, δ)-DP if it satisfies $f_{\epsilon,\delta}$-DP.

Proof. The definition assumes $\alpha \in [0, 1]$, which is the domain of the returned function.

On line 11, arithmetic is computed with upward rounding, so the resulting value is an over-estimate of $\exp(\epsilon)$. On line 12, arithmetic is computed with downward rounding, so the resulting value is an under-estimate of $\exp(-\epsilon)$. The results of these constants are converted exactly into rationals to be used in the tradeoff function.

The function defined on line 14 implements the formula in Definition 1.2 with exact fractional arithmetic, except that it substitutes these conservative rounded constants for $\exp(\epsilon)$ and $\exp(-\epsilon)$.

For $\alpha \in [0, 1]$, the first branch

$$1 - \delta - \texttt{exp_eps} \cdot \alpha$$

is no greater than

$$1 - \delta - \exp(\epsilon)\alpha,$$

because $\texttt{exp_eps} \geq \exp(\epsilon)$ and $\alpha \geq 0$.

For the second branch, if $1 - \delta - \alpha \geq 0$, then

$$\texttt{exp_neg_eps} \cdot (1 - \delta - \alpha) \leq \exp(-\epsilon)(1 - \delta - \alpha),$$

because $\texttt{exp_neg_eps} \leq \exp(-\epsilon)$. If instead $1 - \delta - \alpha < 0$, then both expressions are negative, so they do not affect the outer maximum with 0.

Therefore, for all $\alpha \in [0, 1]$, the function returned by the pseudocode is pointwise no greater than $f_{\epsilon,\delta}(\alpha)$. Hence it is a conservative implementation of the corresponding f-DP tradeoff curve on the stated domain, satisfying the postcondition. □

References

Dong, Jinshuo, Aaron Roth, and Weijie J. Su (2022). "Gaussian Differential Privacy". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 84.1, pp. 3–37. ISSN: 1369-7412.